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Ajgaonkar

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(54) **BLOCKCHAIN RECORDATION AND
VALIDATION OF VIDEO DATA**

(71) Applicant: **Insight Direct USA, Inc.**, Tempe, AZ
(US)

(72) Inventor: **Amol Ajgaonkar**, Chandler, AZ (US)

(73) Assignee: **Insight Direct USA, Inc.**, Chandler, AZ
(US)

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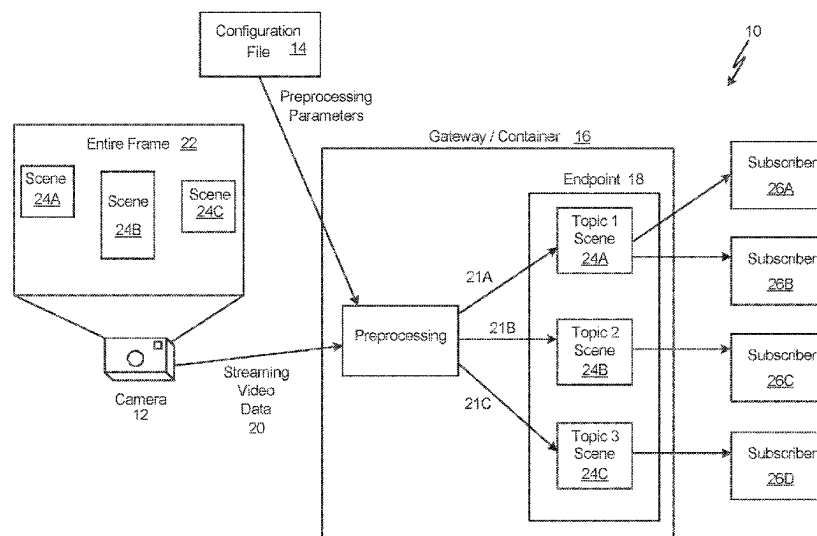
Assistant Examiner — Franklin S Andramuno

(74) *Attorney, Agent, or Firm* — Kinney & Lange, P.A.

(57) **ABSTRACT**

A method of committing video data and associated metadata to a blockchain ledger includes preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model. The method also includes accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data, formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference, and accessing the preprocessed video data and associated metadata by a block-chain subscriber independent from the first artificial intelligence model. In response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, the method includes committing a first entry to the block-chain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data.

20 Claims, 7 Drawing Sheets



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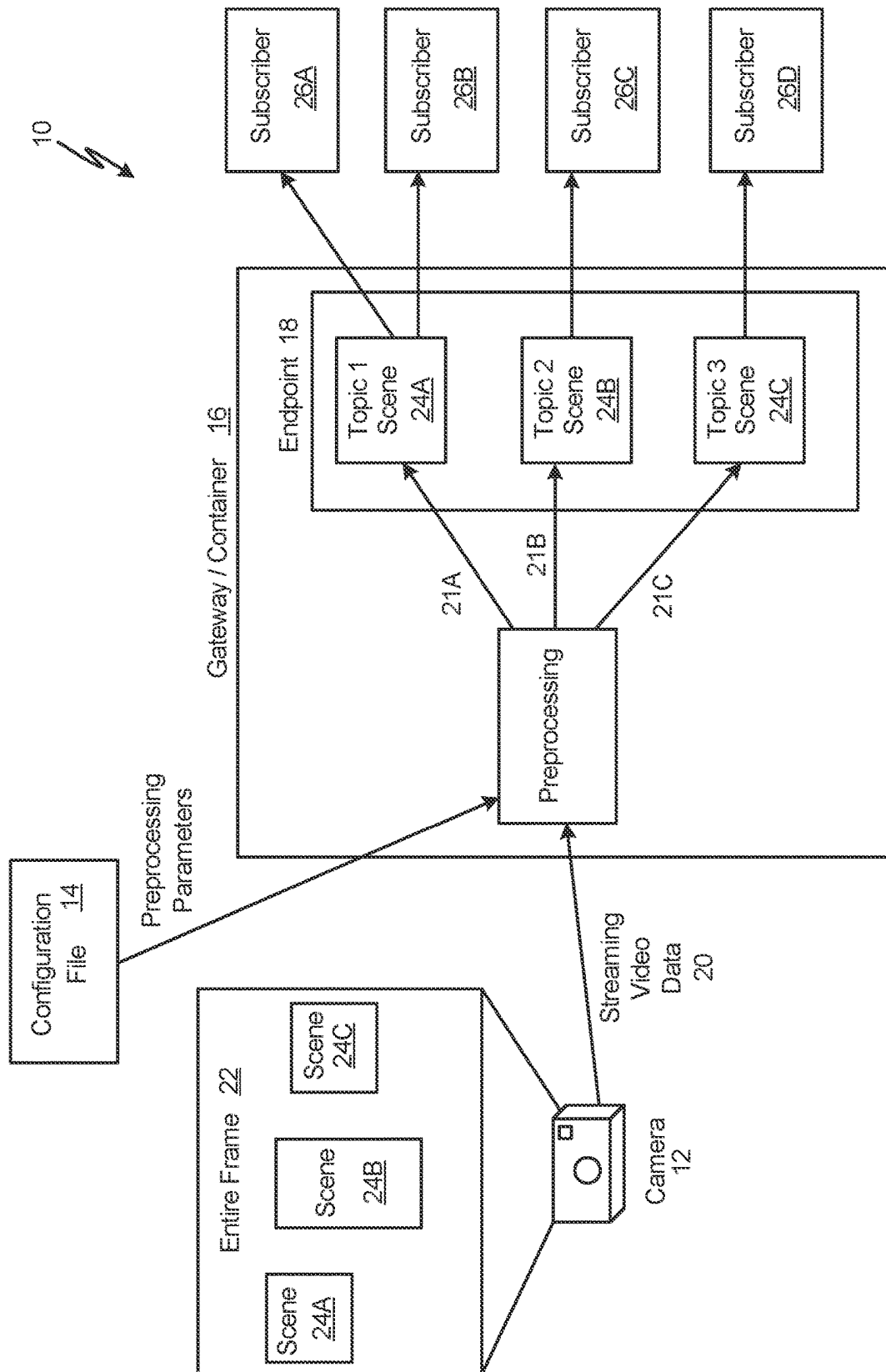


Fig. 1

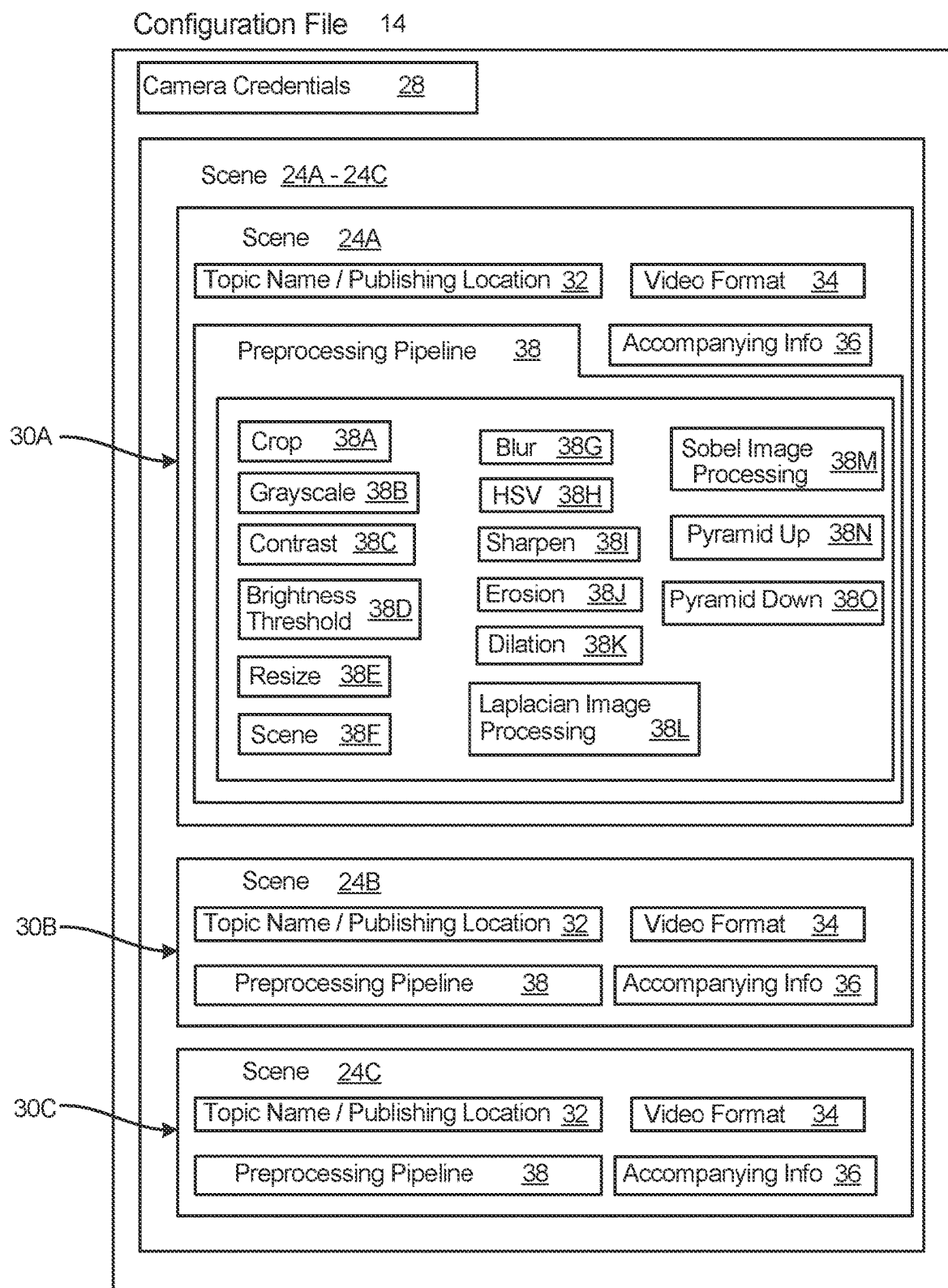


Fig. 2

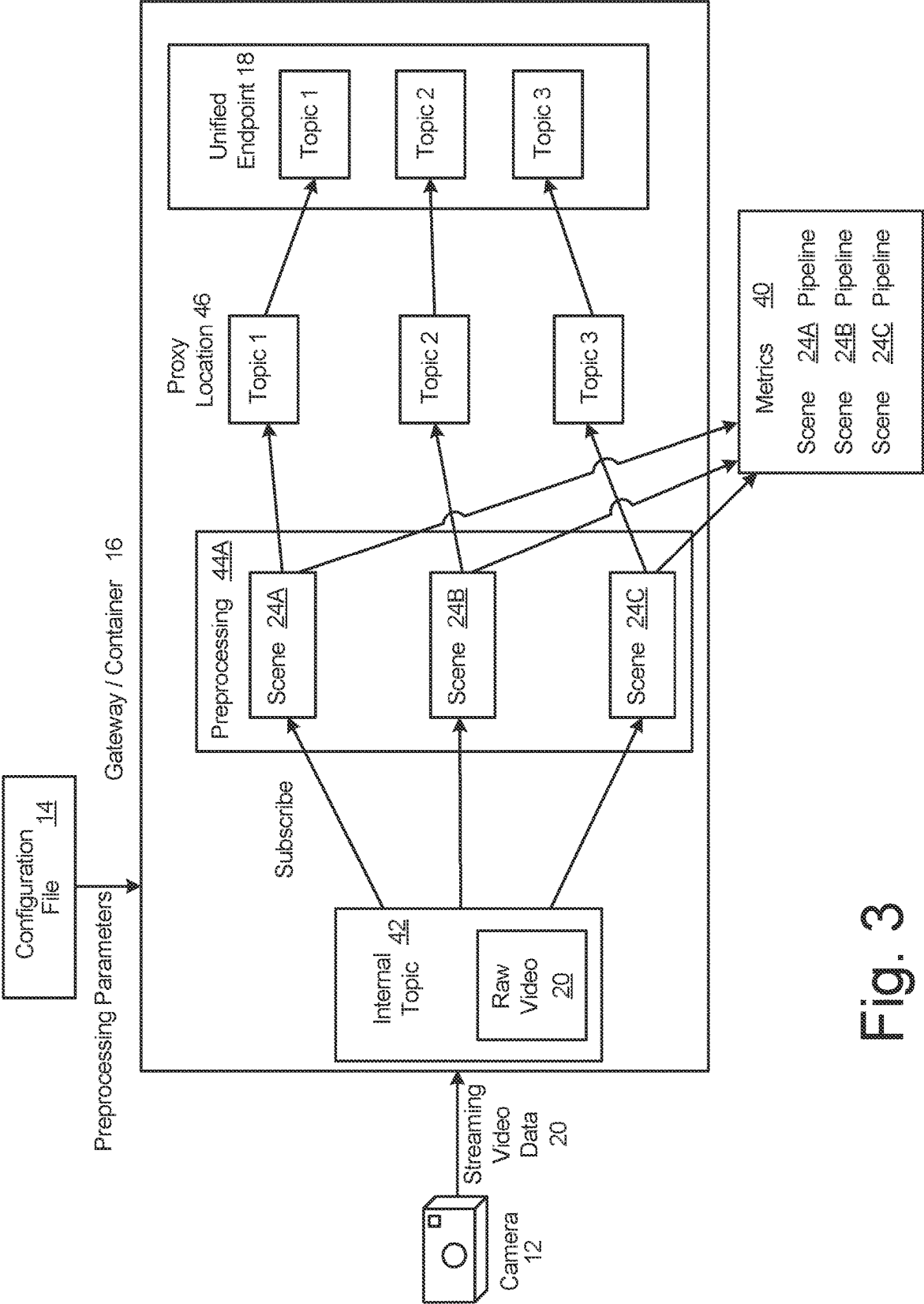


Fig. 3

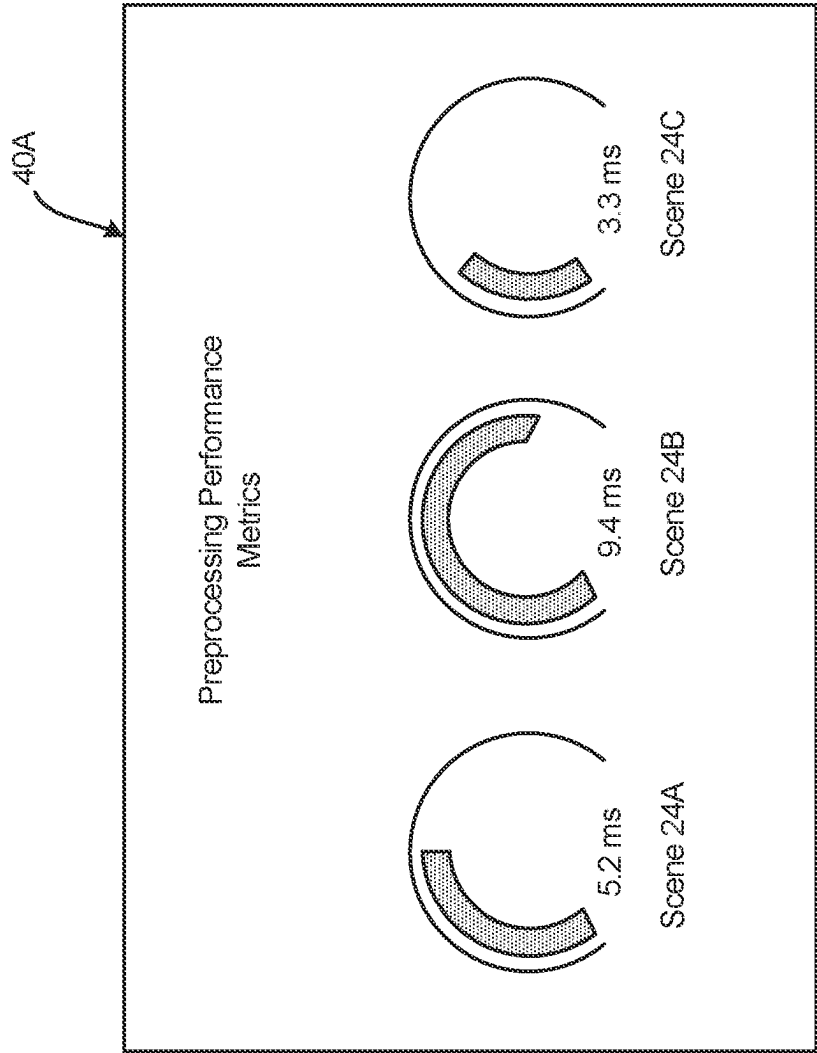


Fig. 4

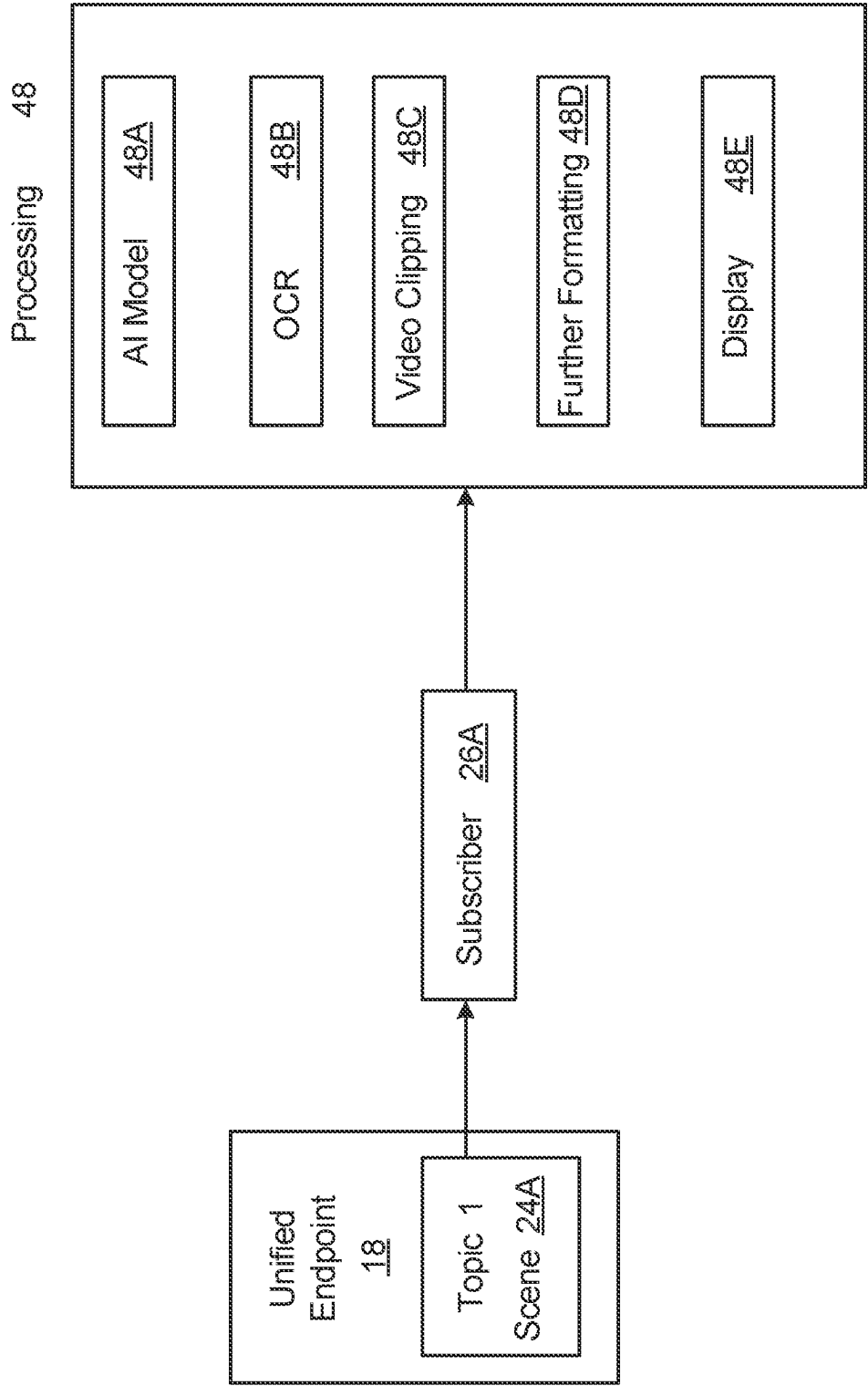


Fig. 5

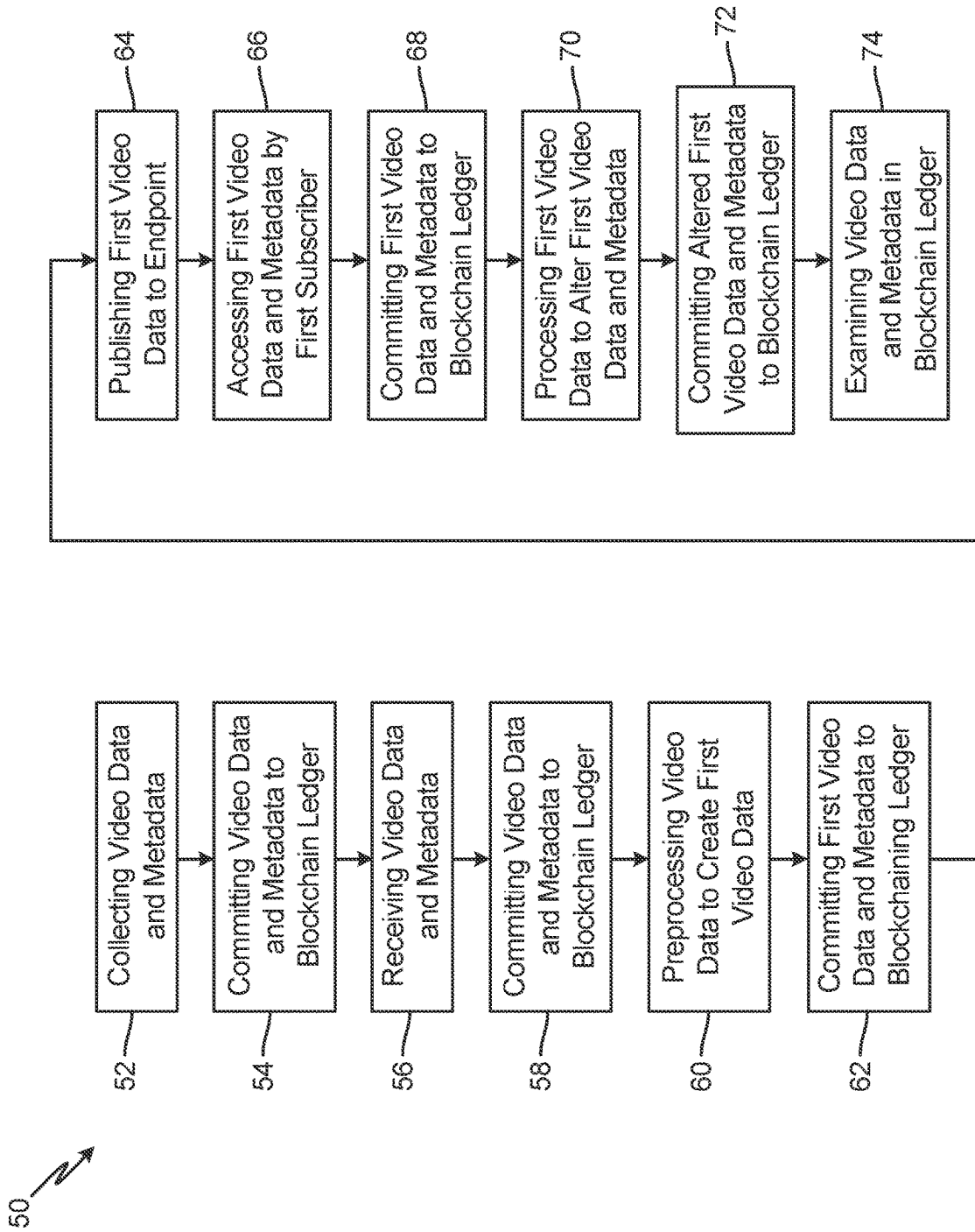


Fig. 6

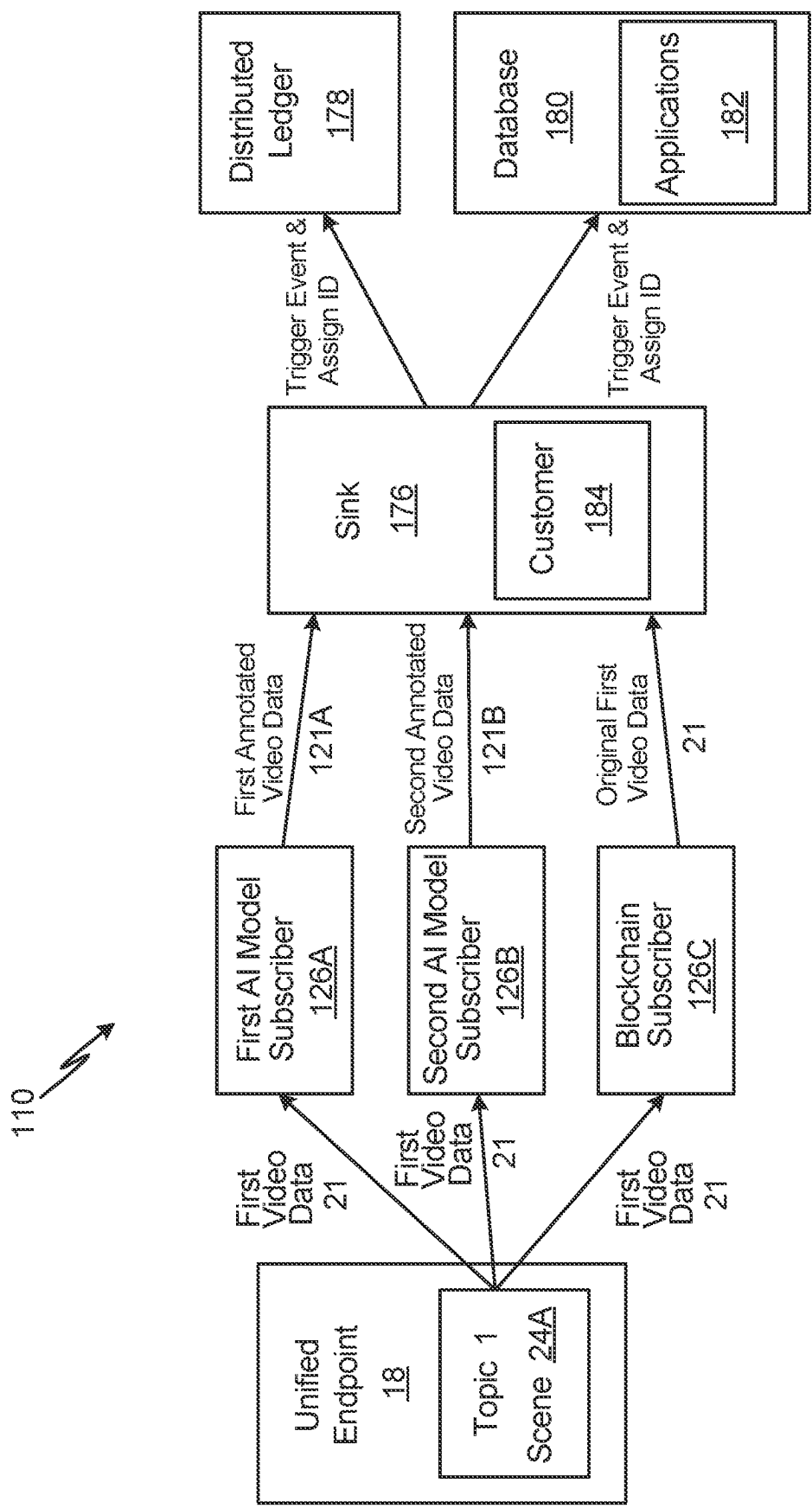


Fig. 7

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**BLOCKCHAIN RECORDATION AND
VALIDATION OF VIDEO DATA****PRIORITY CLAIM**

The present application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/294,554 filed Dec. 29, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

This disclosure relates generally to validating the authenticity of video data and, more specifically, to the use of a blockchain/distributed ledger to record and validate the video data to prevent modification and/or manipulation of the data.

Cameras are beneficial for use in many areas of commercial and personal practice. For example, security cameras are used within (and outside) commercial warehouses and on private personal property. Other applications use cameras along assembly lines for quality control purposes. With the increased capabilities of cameras having higher quality imagery (i.e., resolution) and a wider field of view, more area can be shown in the streaming video by the camera. A large portion of the frame/field of view may be of little or no interest to the consumer (e.g., a security or manufacturing company). However, current practices relay the entirety of the streaming video (i.e., the entire frame/field of view) to the consumer, which can be time and resource consuming due to the need to transfer large frame (i.e., field of view), high resolution video data.

SUMMARY

A system and method for selection/extraction, preprocessing, and publishing of video data of a region of interest (i.e., a scene) that is a subset of a field of view of streaming video is disclosed herein. The system and method can also include processing the video data by a consumer/subscriber after the video data has been published.

Streaming video data is received from a camera with a first field of view. The video data is then preprocessed, by a computer processor such as a gateway or digital container, according to preprocessing parameters defined within a runtime configuration file that is pushed down to the computer processor. The runtime configuration file can be stored and/or edited distant from the computer processor, and any edits/revisions to the runtime configuration file can be pushed to and applied by the computer processor to the streaming video data in real time to alter the preprocessing applied to the video data. The preprocessing can include formatting/cropping the streaming video data received from the camera so as to be first video data of a first region of interest (i.e., a scene) having a second field of view that is less than (shows less area than) the first field of view shown by the entirety of the streaming video data from the camera. The preprocessing as defined by the preprocessing parameters in the runtime configuration file can also include altering the first video data's grayscale, contrast, brightness, color threshold, size, blur, hue saturation value (HSV), sharpen, erosion, dilation, Laplacian image processing, Sobel image processing, pyramid up, and pyramid down (among others). The first video data can then be published to an endpoint (such as a topic on an asynchronous messaging library like ZeroMQ) for subscription and use by a first subscriber/consumer.

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The first video data can then be viewed, used, and/or processed by the first subscriber. The preprocessing as defined in the runtime configuration file can be tailored to the subscriber and the needs/uses of the subscriber and the processing to be performed by the subscriber. For example, the processing performed by the subscriber after publishing of the first video data may be using an artificial intelligence (AI) model to analyze scenarios occurring on/in the first video data. The AI model may require the first video data to be in a particular size, format, etc., which can be selected and applied during the preprocessing as set out in the runtime configuration file so that the subscriber does not need to perform this preprocessing before applying the AI model. The processing, by a computer processor, of the first video data by the subscriber can be performed distant from the camera, the location at which the runtime configuration file is stored and/or edited, and the gateway/container upon which the preprocessing is performed. The first subscriber can perform the processing of the video data to determine at least one output, with the output being indicative of an inference dependent on the first video data. For example, the first video data can be processed by an AI model to determine the amount of a particular product that has passed by on an assembly line (i.e., the amount of the product being an inference dependent on the first video data). The processing can include other operations, such as applying optical character recognition, clipping the first video data to make a video having a specific duration, and/or capturing one frame from the first video data to create a static image of a specific moment of the first video data.

The AI model can determine an output that is indicative of an inference, such as an output that is indicative that a particular product has passed through the field of view shown in the first video data. Whether the inference is correct may come under dispute. For example, a purchaser may dispute whether the manufacturer constructed the correct number of products (as counted by the AI model). In another example, a purchaser may dispute whether the product being constructed was without defects (as determined by the AI model). In such a situation, there may be a need to record the unaltered video data as well as the video data reviewed and potentially annotated by the AI model to provide unalterable, authenticated proof the inferences discovered/determined by the AI model. Recording the video data, annotated video data, and respective associated metadata can be performed by committing the data to a blockchain/distributed ledger, which creates an unalterable entry that can be referenced if the inferences or other information in the video data is disputed. Further entries can be committed to the blockchain ledger, such as an entry that includes video data for each inference that is discovered/determined by the AI model.

One embodiment of a method of committing video data and associated metadata to a blockchain ledger includes preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model. The method also includes accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data, formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference, and accessing the preprocessed video data

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and associated metadata by a blockchain subscriber independent from the first artificial intelligence model. In response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, the method includes committing a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data.

Another embodiment includes a system for authenticating video data and metadata associated with the video data. The system includes a blockchain ledger and a gateway, which includes a first computer processor, that is configured to preprocess the video data according to preprocessing parameters defined within a runtime configuration file to edit the video data to be suitable for use by a first artificial intelligence model. The first artificial intelligence model is configured to access the preprocessed video data and associated metadata, determine a first output that is indicative of a first inference dependent upon the preprocessed video data, and formulate first annotated video data and associated metadata that includes the first output indicative of the first inference. The system also includes a blockchain subscriber, which is executed by a second computer processor, that is independent from the first artificial intelligence model and is configured to access the preprocessed video data and associated metadata and a sink in communication with the first artificial intelligence model and the blockchain subscriber. The sink is configured to, in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, commit a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic of the scene selection and preprocessing system.

FIG. 2 is a schematic of the runtime configuration file.

FIG. 3 is a schematic of a gateway/container along with the inputs and outputs of the gateway/container.

FIG. 4 is a display of the scene preprocessing performance metrics.

FIG. 5 is a schematic of potential processing by a first subscriber.

FIG. 6 is a flow chart showing a method of validating the authenticity of the video data throughout selection, preprocessing, publishing, accessing, and processing of the video data.

FIG. 7 is a schematic of a system for blockchain recording and validation of video data.

While the above-identified figures set forth one or more embodiments of the present disclosure, other embodiments are also contemplated, as noted in the discussion. In all cases, this disclosure presents the invention by way of representation and not limitation. It should be understood that numerous other modifications and embodiments can be devised by those skilled in the art, which fall within the scope and spirit of the principles of the invention. The figures may not be drawn to scale, and applications and embodiments of the present invention may include features and components not specifically shown in the drawings.

DETAILED DESCRIPTION

FIG. 1 is a schematic of system 10 for selection/extraction, preprocessing, and publishing to subscribers of video

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data of a region of interest (i.e., a scene) that is a subset of a first field of view of the streaming video. System 10 can include camera 12, configuration file 14, gateway/container 16, and publishing location/endpoint 18. Camera 12 can include streaming video data 20 having entire frame 22 with a first field of view. Scenes 24A, 24B, and 24C (i.e., individual regions of interest) can be selected/extracted from entire frame 22 each having a second field of view, a third field of view, and a fourth field of view, respectively, that are less than the first field of view of entire frame 22. Camera 12 collects streaming video data 20 and transfers/sends streaming video data 20 to gateway/container 16. Gateway/container 16 preprocesses streaming video data 20 according to preprocessing parameters defined in configuration file 14 and publishes (i.e., allows access/makes available) the preprocessed video data as first video data 21A (for scene 24A), second video data 21B (for scene 24B), and third video data 21C (for scene 24C) to publishing location/endpoint 18. Subscribers 26A-26D can subscribe to video data 21A-21C of each scene 24A-24C located at endpoint 18 to access each scene 24A-24C.

Scene 24A (i.e., a first region of interest) includes first video data 21A, scene 24B (i.e., a second region of interest) includes second video data 21B, and scene 24C (i.e., a third region of interest) includes third video data 21C that are each dependent upon streaming video data 20. In one example, first video data 21A forming scene 24A (i.e., a first region of interest) has a second field of view that is less than the first field of view comprising entire frame 22 shown by camera 12 (as streaming video data 20).

System 10 can include machine-readable storage media. In some examples, a machine-readable storage medium can include a non-transitory medium. The term “non-transitory” can indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium can store data that can, over time, change (e.g., in RAM or cache). In some examples, storage media can be entirely or in part a temporary memory, meaning that a primary purpose storage media is not long-term storage. Storage media, in some examples, is described as volatile memory, meaning that the memory, does not maintain stored contents when power to system 10 (or the component(s) where storage media are located) is turned off. Examples of volatile memories can include random access memories (RAM), dynamic random-access memories (DRAM), static random-access memories (SRAM), and other forms of volatile memories. In some examples, storage media can also include one or more machine-readable storage media. Storage media can be configured to store larger amounts of information than volatile memory. Storage media can further be configured for long-term storage of information. In some examples, storage media include non-volatile storage elements. Examples of such non-volatile storage elements can include magnetic hard discs, optical discs, flash memories and other forms of solid-state memory, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories. Most generally, storage media is machine-readable data storage capable of housing stored data from a stored data archive.

System 10 can also include one or multiple computer/data processors. In general, the computer/data processors can include any or more than one of a processor, a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or other equivalent discrete or integrated logic circuitry. System 10 can include other

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components not expressly disclosed herein but that are suitable for performing the functions of system 10 and associated methods of preprocessing and processing video data and other forms of data. For example, system 10 can include communication software and/or hardware for pushing/sending configuration file 14 from storage media at a location distant from gateway/container 16, such as cloud storage, to gateway/container 16 for execution/implementation on streaming video data 20.

Camera 12 can be any device capable of collecting streaming video data 20, such as a Real Time Streaming Protocol (RTSP) camera or a USB camera. Streaming video data 20 can be video data that is continuously captured/recorded by camera 12 in any suitable format. Camera 12 can be positioned/located to provide streaming video data 20 displaying entire frame 22 with a first field of view. The first field of view shown/displayed by camera 12 can be a wide field of view that shows multiple regions of interest. Video data 20 being collected, for example, can show a wide field of view of a warehouse for storing commercial products and/or an assembly line producing commercial products of which each individual subscriber 26A-26D may only be interested in a region/scene 24A-24C that is a subset of entire frame 22. Camera 12 can collect and transfer streaming video data 20 in any resolution/video quality and any format, including (but not limited to) MP4, AVI, FLV, WMV, MOV, MPEG, Motion JPEG, AVCHD, WebM, and/or MKV. Camera 12 can transfer/send streaming video data 20 to gateway/container 16 over any suitable means, including via the internet, short-range wireless technology, or any other type of wired and/or wireless connection.

Configuration file 14 is an editable file that contains preprocessing parameters that define, among other instructions, how streaming video data 20 is to be preprocessed by gateway/container 16 to create video data 21A-21C. Configuration file 14 can include numerous other instructions for gateway/container 16, including which camera 12 to connect to (i.e., receive streaming video data 20 from), what portion of entire frame 22 to select/extract to create scenes 24A-24C (i.e., how to crop frame 22 to create scenes 24A-24C), and at which endpoint 18 to publish the preprocessed scenes 24A-24C. This is discussed in detail with regards to FIG. 2. Configuration file 14 can be edited/revised and pushed/conveyed to gateway/container 16 for execution in real time (i.e., runtime) such that an editor can revise the preprocessing parameters and those revisions can be applied to scenes 24A-24C at runtime. Configuration file 14 can be an executable program file or have another format for including instructions and conveying information that is then used by gateway/container 16 to apply the preprocessing to video streaming data 20. Additionally, configuration file 14 can be stored in storage media adjacent to and/or part of gateway/container 16 or in storage media distant from gateway/container 16, such as in the cloud. Configuration file 14 can be accessible only by one editor or can be accessible by multiple parties (e.g., editors), which may include subscribers 26A-26D who can edit the preprocessing parameters defined by/within configuration file 14 to instruct gateway/container 16 to preprocess one or each of scenes 24A-24C depending on the needs/desires of subscribers 26A-26D, respectively.

Gateway/container 16 can include a computer processor capable of performing instructions provided by configuration file 14, which can include preprocessing parameters that are to be applied to streaming video data 20. Gateway/container 16 can be a gateway node, edge device, container, virtual machine, or other software and/or hardware able to

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accept configuration file 14 and perform the instructions therein to apply the preprocessing parameters to streaming video data 20. Further, gateway/container 16 can be within a single computer hardware set up due to virtualization. Gateway/container 16 can include one or multiple storage media for storing information, such as the preprocessing parameters pushed/sent to gateway/container 16 by/from configuration file 14 and/or other information like streaming video data 20. Gateway/container 16 can be located at various locations, including adjacent to camera 12 and/or on the same network as camera 12, distant from camera 12 with streaming video data 20 being received by gateway/container 16 from camera 12 via a wired or wireless connection, in the cloud, or at multiple locations. Gateway/container 16 is in communication with configuration file 14 to accept instructions for applying preprocessing parameters. Additionally, gateway/container 16 may be configured to contact configuration file 14 to determine if configuration file 14 has been edited. If edited, gateway/container 16 can perform preprocessing (on streaming video data 20 being received) according to the newly edited configuration file 14. In other examples, gateway/container 16 can utilize preprocessing parameters or other information included within configuration file 14 on a period basis, and can utilize changes (or edits) made to configuration file 14 when encountered in a next iteration of the period.

For clarity, streaming video data 20 is unaltered video data that is received by gateway/container 16 from camera 12, whereas first video data 21A is video data that has been preprocessed by gateway/container 16 (according to preprocessing parameters defined in configuration file 14) to create scene 24A. Similarly, second video data 21B is video data that has been preprocessed to create scene 24B, and third video data 21C is video data that has been preprocessed to create scene 24C. For each of first, second, and third video data 21A-21C, the originating video data is streaming video data 20 (which can be converted to raw video data 20A as described with regards to FIG. 3 below). Each of scenes 24A-24C can be a subset of entire frame 22 and show second, third, and fourth fields of view, respectively, that are less than the first field of view of entire frame 22. However, other scenes can have the same field of view as first field of view of entire frame 22 and instead, other editing is performed on that scene besides cropping; for example, that scene can be edited to be in grayscale whereas entire frame 22 of streaming video data 20 is in color.

Gateway/container 16 can be in communication with endpoint 18 to which gateway/container 16 publishes the preprocessed video data 21A-21C (e.g., scenes 24A-24C). The communication can be wired or wireless, such as communication via the internet. However, endpoint 18 can be at the same location as gateway/container 16 or on the same computer hardware set up and/or network. Further, endpoint 18 can be located on the internet with a unique address and/or security protocol that allows for subscription and access to scenes 24A, 24B, and 24C. Scenes 24A, 24B, and 24C can be published to endpoint 18 using an asynchronous messaging library, for example ZeroMQ, such that scenes 24A, 24B, and 24C are published as topic 1, topic 2, and topic 3, respectively. Subscribers 26A-26D can subscribe to any of topics 1-3 to receive video data 21A-21C of scenes 24A, 24B, 24C, respectively. Gateway/container 16 can publish video data 21A-21C of each of scenes 24A-24C to endpoint 18 in any format suitable for use by subscribers 26A-26D. For example, video data 21A-21C can each be published as Motion JPEG or any of the formats listed above with regards to streaming video data 20. The format that

video data 21A-21C of each of scenes 24A-24C can be designated in configuration file 14 and applied to video data 21A-21C by gateway/container 16.

Each scene 24A-24C at topics 1-3, respectively, can be subscribed to by any number of subscribers 26A-26D. In the example shown in FIG. 1, scene 24A has two subscribers 26A and 26B, while scene 24B has one subscriber 26C, and scene 24C has one subscriber 26D. Video data 21A-21C of each of scenes 24A-24C can be further processed by subscriber 26A-26D, respectively, depending on the desired output/inference to be determined from video data 21A-21C. For example, first video data 21A of scene 24A can be further processed by an AI model to determine the amount of a particular product that has passed by camera 12 (in the second field of view of scene 24A) on an assembly line. This is described in detail with regards to FIG. 5.

FIG. 2 is a schematic of configuration file 14. Each "component" of configuration file 14 can be one or a set of instructions that, when executed by gateway/container 16, applies a process/edit. Thus, while this disclosure may discuss the components of configuration file 14 as being physical, tangible elements, the components can be one or multiple groups of executable software code contained within configuration file 14. Configuration file 14 includes information regarding camera credentials 28 and preprocessing parameters 30A-30C corresponding to scenes 24A-24C. Preprocessing parameters 30B and 30C can include the same type of information/parameters as preprocessing parameters 30A. However, for simplicity, the possibilities of information/parameters shown in FIG. 2 with regards to preprocessing parameters 30A are not shown for preprocessing parameters 30B and 30C. However, preprocessing parameters 30B and 30C can be the same or different from each other and from preprocessing parameters 30A corresponding to scene 24A. Preprocessing parameters 30A can include topic name/publishing location 32, video format 34, accompanying information 36, and preprocessing pipeline 38 having various video edits 38A-38O. Video edits 38A-38O can include the nonexclusive list of crop 38A, grayscale 38B, contrast 38C, brightness 38D, threshold 38E, resize 38F, blur 38G, hue saturation value (HSV) 38H, sharpen 38I, erosion 38J, dilation 38K, Laplacian image processing 38L, Sobel image processing 38M, pyramid up 38N, and pyramid down 38O.

Configuration file 14 can be edited and pushed/conveyed to gateway/container 16 in real time (i.e., runtime) such that preprocessing parameters 30A-30C (and the other information contained in configuration file 14) can be applied to streaming video data 20 immediately to preprocess and output video data 21A-21C. Configuration file 14 can be stored, edited, and/or pushed/conveyed to gateway/container 16 in any suitable format/file type, such as a text file, a comma separated value file, or other format/file type. Configuration file 14 can include other information/parameters not expressly disclosed herein and not shown in FIG. 2.

Configuration file 14 can include camera credentials 28, which provides the information needed for gateway/container 16 to connect to camera 12 and/or receive streaming video data 20 from camera 12. Camera credentials 28 can include other information such as encryption/decryption information, security access information, and/or instructions for beginning and/or ending the collection of streaming video data 20 by camera 12. Camera credentials 28 can include information for connecting to multiple cameras 12 and/or information for gateway/container 16 to receive the same or different streaming video data 20 from the same or different cameras 12 for different scenes 24A-24C. In one

example, camera credentials 28 are provided once and applied to all scenes 24A-24C. In another example, different camera credentials 28 are provided for each scene 24A-24C and applied to each of scenes 24A-24C individually.

Configuration file 14 also includes information specific to each scene 24A-24C. The information/instructions are designated as preprocessing parameters 30A, 30B, and 30C, respectively. Preprocessing parameters 30A-30C are used by gateway/container 16 (e.g., in the form of executable instructions or indications of the executable instructions) and applied to streaming video data 20 to create video data 21A-21C of scenes 24A-24C, respectively. Preprocessing parameters 30A-30C can include topic name/publishing location 32, which designates where video data 21A-21C of scenes 24A-24C will be published after preprocessing. As described below with regards to gateway/container 16 and FIG. 3, publishing location 32 can be a proxy location which is then relayed to unified endpoint 18 to make locating the topic/scenes 24A-24C easier for subscribers 26A-26D. Publishing location 32 can be any other location suitable for providing access to subscribers 26A-26D.

Configuration file 14 can designate video format 34 that each of scenes 24A-24C is to be published at, which can be the same format as streaming video data 20 or any other type of suitable video format, including the formats listed above with regards to streaming video data 20 and/or video data 21A-21C. Scenes 24A-24C can be published having the same video format 34 or different video formats 34. If the format of the video data of scenes 24A-24C is to be changed, the steps of changing the video format can be performed before, during, or after any of the other instructions/steps set out in preprocessing parameters 30A-30C. For example, video format 34 can be changed before, during, or after video edits 38A-38O are performed by gateway/container 16.

Preprocessing parameters 30A-30C can also include accompanying information 36, which is information provided/published with video data 21A-21C for each of scenes 24A-24C. Accompanying information 36 can include any information about first video data 21 that may be of use to subscribers 26A-26C. For example, accompanying information 36 can include first video data 21A frame size, which may be helpful in indicating to subscriber 26A what processing should be performed on first video data 21A of scene 24A; if the frame size is 720 pixels by 486 lines, first video data 21A of scene 24A may be most suitable for processing by an AI model. Accompanying information 36 can include metadata and/or other information regarding what preprocessing has been performed on streaming video data 20 to create video data 21A-21C for scenes 24A-24C, respectively.

Preprocessing parameters 30A-30C can also include preprocessing pipeline 38 that includes numerous video edits 38A-38O that can be applied to streaming video data 20 to create video data 21A-21C for each of scenes 24A-24C. Preprocessing pipeline 38 can designate the instructions for the entire video edits made to streaming video data 20 for each of scenes 24A-24C. The amount of time gateway/container 16 takes to perform the video edits designated by preprocessing pipeline 38 (i.e., video edits 38A-38O) for each of scenes 24A-24C can be measured, recorded, and displayed as preprocessing performance metrics (see FIG. 4). The order in which video edits 38A-38O are performed can be optimized (or otherwise enhanced for increasing accuracy, decreasing bandwidth or latency, etc.) by gateway/container 16 and/or optimized within configuration file 14 to reduce the amount of time gateway/container 16 takes to

perform preprocessing pipeline **38**. This optimization can be done manually by an editor (e.g., the individual with access to edit configuration file **14**) or automatically by system **10**. For example, depending on which video edits **38A-380** are to be performed in preprocessing pipeline **38** by gateway/container **16**, the order of performance of those video edits **38A-380** can be rearranged to reduce the amount of time gateway/container **16** takes to perform preprocessing pipeline **38**. In one example, crop **38A** is the first video edit **38A-380** to be performed, followed by other video edits **38B-380**.

Video edits **38A-380** are a nonexclusive list of edits that can be designated in configuration file **14** and performed on streaming video data **20** by gateway/container **16**. Preprocessing pipeline **38** can include other video edits not expressly included in the list of video edits **38A-380**. Similarly, not all of video edits **38A-380** need to be performed to create video data **21A-21C** of each of scenes **24A-24C**, and different scenes **24A-24C** can include different video edits **38A-380** performed on streaming video data **20** by gateway/container **16**. In one example, only crop **38A** is performed on streaming video data **20** to create first video data **21A** of scene **24A**, while a different crop **38A** and brightness **38D**, resize **38E**, and dilation **38K** are performed on streaming video data **20** to create second video data **21B** of scene **24B** that is different than first video data **21A** of scene **24A**.

Each of video edits **38A-380** are briefly described as follows. Crop **38A** is the removal of unnecessary areas/regions (i.e., regions that are not of-interest to the subscriber) of entire frame **22** having first field of view to create scenes **24A-24C** each with second, third, and fourth field of views, respectively. Scenes **24A-24C** that have been cropped **38A** have fields of view that are a subset of (i.e., less than) first field of view of entire frame **22**. Grayscale **38B** is the alteration of the color of video data **21A-21C** and can include limiting the color to be between white and black. Contrast **38C** is the alteration of the difference between the maximum and minimum pixel intensity. Brightness **38D** is the alteration in the luminous brightness of video data **21A-21C**. Threshold **38E** is the alteration of the color of video data **21A-21C** by changing the color of select pixels of video data **21A-21C** that are above a specified threshold color value. Resize **38F** is the alteration of the frame size of video data **21A-21C** without cutting/cropping any of the frame out. Blur **38G** is the alteration of the clarity of video data **21A-21C**, which may be desired for some processing applications, such as an AI model, performed by subscribers **26A-26D**. Hue saturation value (HSV) **38H** is assigning a numerical readout of video data **21A-21C** that corresponds to the color contained therein. Sharpen **38I** is altering video data **21A-21C** to make the objects therein appear more defined/sharpened. Erosion **38J** is altering video data **21A-21C** by shrinking pixels and/or removing pixels on object boundaries, while dilation **38K** is the reverse of erosion in that video data **21A-21C** is enlarged by resizing pixels and/or adding pixels at object boundaries. Laplacian image processing **38L** and Sobel image processing **38M** are processing techniques known in the art that can be applied to video data **21A-21C**. Pyramid up **38N** and pyramid down **380** are altering video data **21A-21C** by smoothing and/or subsampling as known in the art. Each of scenes **24A-24C** can include these and other video edits **38A-380** to be applied by gateway/container **16** to preprocess streaming video data **20** and output as scenes **24A-24C** for use by subscribers **26A-26D**.

Configuration file **14** can arrange the instructions of camera credentials **28** and preprocessing parameters **30A-**

30C to be performed in any order, or gateway/container **16** can have the capabilities to arrange/rearrange the information/instructions to be performed in a desired/optimized sequence. Additionally, gateway/container **16** can be configured to perform each set of preprocessing parameters **30A-30C** in parallel such that preprocessing parameters **30A**, preprocessing parameters **30B**, and preprocessing parameters **30C** are performed at the same time (and the time gateway/container **16** takes to perform those preprocessing parameters are measured, recorded, and displayed as metrics). Configuration file **14** can be edited at any time by an editor and then pushed/conveyed/accessed by gateway/container **16** at runtime such that the preprocessing of video data **21A-21C** is altered according to the newly edited configuration file **14** at runtime.

FIG. 3 is a schematic of gateway/container **16** (hereinafter, “gateway **16**”) along with inputs to and outputs from gateway **16**. Each “component” of gateway **16** (and corresponding inputs and outputs) can be one or a set of instructions, programs, processors, storage media locations, and/or other software or hardware used to select/extract, preprocess, and publish video data **21A-21C** as scenes **24A-24C**. Thus, while this disclosure may discuss the components of gateway **16** (and corresponding inputs and output) as being physical, tangible elements, the components can be partially or entirely contained within software and/or hardware.

Inputs to gateway **16** can be streaming video data **20** (received from camera **12**) and configuration file **14**, which includes camera credentials **28** and preprocessing parameters **30A-30C**. Outputs from gateway **16** can be scenes **24A-24C** to unified endpoint **18**, which is the location at which scenes **24A-24C** are published as topics **1-3**, and metrics **40**, which is the location at which preprocessing pipeline **38** information (i.e., the amount of time gateway **16** takes to apply preprocessing parameters **30A-30C** to streaming video data **20** to create scenes **24A-24C**) is published/accessible. Gateway **16** can include raw video data **20A**, which is streaming video data **20** that has been published at internal topic **42** and to which gateway **16** subscribes to receive video data used to create each of scenes **24A-24C**. Gateway **16** can include preprocessing pipeline optimization and preprocessing **44**, which uses preprocessing parameters **30A-30C** as defined in configuration file **14** to preprocess streaming video data **20** (accessed as raw video data **20A** at internal topic **42**) to create scenes **24A-24C**. Gateway **16** publishes scenes **24A-24C** to proxy location **46** at topics **1-3**, respectively. Gateway **16** can then publish/relay scenes **24A-24C** (having video data **21A-21C**) from proxy location **46** to unified endpoint **18**.

Gateway **16** receives streaming video data **20** from camera **12** and can publish the video data as raw video data **20A** at internal topic **42**. This configuration provides for a constant, known location of published raw video data **20A** independent of where the original streaming video data **20** is received from. Thus, if the location where streaming video data **20** is being received from changes (e.g., if one camera is disconnected but another camera comes online), raw video **20A** will still be accessible at internal topic **42** without the need to change where gateway **16** is looking for video data to create scenes **24A-24C**, thus ensuring a smooth transition of incoming streaming video data **20**. Raw video data **20A** published at internal topic **42** can be configured such that only gateway **16** has access.

Gateway **16** can be provided with the information in configuration file **14** (i.e., camera credentials **28** and preprocessing parameters **30A-30C**) via a variety of avenues. In one example, gateway **16** has location information of con-

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figuration file 14 and actively accesses configuration file 14. In another example, configuration file 14 is pushed/conveyed to gateway 16 once, periodically, or continuously and gateway 16 passively waits to begin preprocessing streaming video data 20 until configuration file 14 has been received. Another example can be a combination of the two above examples in that gateway 16 actively accesses configuration file 14 at the beginning of preprocessing (and continues preprocessing following those known instructions) and configuration file 14 is pushed/conveyed to gateway 16 only after configuration file 14 has been edited/revised.

Gateway 16 can perform pipeline optimization on preprocessing parameters 30A-30C. As discussed above, pipeline optimization can be performed by gateway 16 (or another component) to reduce the time gateway 16 takes to preprocess raw video data 20A to create video data 21A-21C (and attach any additional information) of scenes 24A-24C and/or to increase accuracy of the preprocessing operations. Pipeline optimization can include arranging/rearranging the order in which video edits 38A-380 are performed by gateway 16.

Gateway 16 can then, according to preprocessing parameters 30A-30C defined within configuration file 14, preprocess 44 raw video data 20A (which is derived from and similar to streaming video data 20) to edit raw video data 20A to create video data 21A-21C of scenes 24A-24C. The preprocessing to create each of scenes 24A-24C can be performed individually for each scene and can be performed in parallel (i.e., simultaneously). The preprocessing performed by gateway 16 can edit each of scenes 24A-24C to the desires/needs of subscribers 26A-26C, respectively. For example, gateway 16 can preprocess raw video data 20A to crop a first field of view of entire frame 22 to eliminate areas/regions of the first field of view that are of no interest to subscriber 26A and keep a second field of view of, for example, scene 24A that is a subset of the first field of view. Thus, further processing by subscriber 26A (or subscriber 26B) does not need to be performed on the areas (regions not of interest) eliminated/trimmed by the cropping performed during preprocessing. In this example, the cropping reduces the processing time and resources needed by subscriber 26A. The alteration of video format can be included in the preprocessing and/or can be performed before, during, or after the application of other preprocessing parameters 30A-30C.

Scenes 24A-24C, which are made up of raw video data 20A that has been preprocessed according to configuration file 14 to create video data 21A-21C, are published/sent to proxy location 44. Because scenes 24A-24C are continuous video data, scenes 24A-24C are continuously published (e.g., made available) to subscribers 26A-26C, respectively. Scenes 24A-24C are published/sent first to proxy location 44 and then relayed to unified endpoint 18. Having scenes 24A-24C first being published/sent to proxy location 44 ensures that, no matter what path scenes 24A-24C take, video data 21A-21C of scenes 24A-24C will always end up at proxy location 44 and then be relayed to unified endpoint 18. Because scenes 24A-24C always end up at proxy location 44, unified endpoint 18 always knows the location to access scenes 24A-24C and can actively look to proxy location 44 to obtain scenes 24A-24C or passively wait for scenes 24A-24C to be published/sent to unified endpoint 18.

During preprocessing to create scenes 24A-24C, the amount of time gateway 16 takes to apply all of preprocessing parameters 30A-30C to raw video data 20A to create scenes 24A-24C, respectively, (or the amount of time gate-

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way 16 takes to apply only video edits 38A-380 of preprocessing pipeline 38, depending on the desired measurement) is measured, recorded, and published at metrics 40 for viewing by an editor or any party with access. Metrics 40 can be published on an asynchronous messaging library like ZeroMQ (similar to the publishing of scenes 24A-24C) or can be displayed on a user interface similar to FIG. 4, which shows a display of scene preprocessing performance metrics 40A. Metrics 40 can be outputted from gateway 16 to numerous other systems and/or locations, including to the cloud or another location distant from gateway 16.

As shown in FIG. 4, scene preprocessing performance metrics 40A show a value of time for each of scenes 24A-24C. This value of time displayed is the amount of time gateway 16 takes to apply preprocessing parameters 30A-30C for each of scenes 24A-24C. Alternatively, this value of time displayed can be the amount of time gateway 16 takes to apply only video edits 38A-380 of preprocessing pipeline 38 and not the entirety of preprocessing parameters 30A-30C. This measurement may be desired because the other instructions/information of preprocessing parameters 30A-30C (other than video edits 38A-380) may not change between scenes 24A-24C while preprocessing pipeline 38 (the video edits 38A-380 being applied) may be drastically different from scene to scene and thus the amount of time for performing preprocessing pipeline 38 may vary greatly from scene to scene.

For example, preprocessing performance metrics 40A in FIG. 4 show an amount of time gateway 16 takes to perform the video edits of preprocessing pipeline 38 for scene 24A as 5.2 milliseconds (ms), for scene 24B as 9.4 ms, and for scene 24C as 3.3 ms. In this example, it may be desirable or necessary to alter preprocessing parameters 30B in configuration file 14 (either remove some edit and/or rearrange the order in which those edits are performed by gateway 16) to reduce the amount of time gateway 16 takes to apply preprocessing pipeline 38 to create/alter scene 24B. As discussed above, gateway 16 can have the capability to optimize the order in which the edits/instructions in preprocessing parameters 30A-30C and/or preprocessing pipelines 38 are performed to reduce the amount of time needed to preprocess/apply the edits/instructions. Thus, preprocessing performance metrics 40A as shown in FIG. 4 may be the shortest amount of time gateway 16 takes to perform preprocessing parameters 30A-30C (or preprocessing pipeline 38, depending on the chosen measuring points).

Scenes 24A-24C can be published from proxy location 44 to unified endpoint 18 as topics 1-3 within an asynchronous messaging library, such as ZeroMQ. Unified endpoint 18 allows for a known, constant location to which subscribers 26A-26C can look to access scenes 24A-24C, respectively. If more scenes are created, those scenes would be published and accessible at unified endpoint 18, so subscribers 26A-26D and/or other subscribers would know where to look to access the additional scenes.

FIG. 5 is a schematic showing processing 48 capable of being performed on scene 24A by first subscriber 26A. Processing 48 can also be performed on scenes 24B and 24C or other scenes not disclosed herein.

After being published/sent to endpoint 18, scenes 24A-24C are available to be subscribed to and accessed by subscribers 26A-26D. Each scene can be subscribed to by any number of subscribers as is necessary/desirable. For example, a scene may be subscribed to by numerous subscribers each running processing 48 that includes different AI models. In this example, one AI model can be determining the amount of a first product that is passing through the

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scene on an assembly line, while a second AI model can be determining the amount of a second product that is passing through the scene on the same assembly line. In this case, the scene is unchanged between the two AI models (i.e., between the two subscribers) but the processing performed by each subscriber after the scene has been published is different. It should be noted that one entity (e.g., a person, company, quality control sector) can subscribe to a scene multiple times and thus constitute multiple subscribers. As discussed above, the preprocessing performed to create/alter each scene can be tailored to the needs of the subscriber(s) to, for example, reduce processing 48 resources and time needed to determine at least one output that is indicative of an inference the subscriber is aiming to ascertain.

The example in FIG. 5 shows first subscriber 26A performing processing 48 on scene 24A. Processing 48, as selected and executed by subscriber 26A (either automatically and/or manually by a computer processor and/or other hardware and software), can include AI model 48A, optical character recognition (OCR) 48B, video clipping 48C, further formatting 48D, and display 48E of the video data of scene 24A. Processing 48 can include other instructions/edits not expressly disclosed in FIG. 5 and listed above.

The disclosed potential instructions/edits that subscriber 26A can perform in processing 48 are as follows. AI model 48A can be a program/model that may have machine learning and can use scene 24A to determine at least one output indicative of an inference dependent upon scene 24A. The inference, for example, can be the amount of a specific product that is viewable in scene 24A over a defined period of time. AI model 48A can also be, for example, a program/model that determines how many people appear in scene 24A over a defined period of time. AI model 48A can include other capabilities and/or configurations. OCR 48B can be a program (or other configuration) that recognizes and records any characters (i.e., text) that appear in scene 24A. For example, scene 24A can be video data of a street and OCR 48B will recognize and record any text that appears on the side of a vehicle, such as a delivery truck, that is in scene 24A. Video clipping 48C can clip the video data shown in scene 24A to create a clip of a defined period of time, and/or video clipping 48C can clip the video data shown in scene 24A to create a static image of a defined moment in time. Further formatting 48D can be video edits, such as video edits 38A-380 in configuration file 14, or any other video or file formatting that are performed by subscriber 26A. For example, further formatting 48D can include cropping scene 24A to be a subset of the second field of view shown in scene 24A. Display 48E can be making scene 24A viewable on a screen or other visual display. Display 48E can also include any video formatting/reconfiguring that is necessary to effectuate the display of scene 24A. While video edits 38A-380 and further formatting 48D to create and/or process video data 21A-21C have included only edits to the video/image, edits can include editing the audio or other aspects of the video data.

The potential instructions/edits (i.e., processing 48) can be performed in parallel or series. Further, processing 48 can be configured such that instructions/edits 48A-48E work together such that one instruction/edit is prompted by an inference from another instruction/edit. For example, video clipping 48C can be configured to work in tandem with another edit/process; if AI model 48A determines that a product is defective, video clipping 48C can be prompted to record and clip a particular duration (or moment to create a static image) of scene 24A showing the defective product and save the clip (or image) for proof/validation.

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System 10, with associated methods, for selection/extraction, preprocessing, and publishing of streaming video data 20 into scenes 24A-24C and for processing scenes 24A-24C is used to determine at least one output that is indicative of an inference dependent upon video data 21A-21C. System 10 reduces the preprocessing/processing time and resources necessary for accomplishing the desired output/determining the selected inference. System 10 allows for a streamlined process that extends from collecting streaming video data 20 from camera 12 to making preprocessing scenes 24A-24C available to subscribers 26A-26C for further processing and analysis/determinations. System 10 also allows for measurement, recordation, and viewing of preprocessing performance metrics 40 and optimization of preprocessing pipeline 38 (and/or preprocessing parameters 30A-30C) to reduce the amount of time and resources needed to apply preprocessing parameters 30A-30C to streaming video data 20 (i.e., raw video data 20A) to create video data 21A-21C of scenes 24A-24C.

FIG. 6 shows a flow chart of method 50 for validating the authenticity of video data throughout selection, preprocessing, publishing, accessing, and processing of the video data. Throughout method 50, the video data after each step is committed to a blockchain ledger to record and authenticate the video data to ensure the video data has not been manipulated and to keep a record of the state/properties of the video data after receiving, preprocessing, publishing, accessing, and processing the video data. Committing the video data to the blockchain ledger, with validation by multiple nodes of the distributed ledger, after each step and/or after the video data has changed hands (from one computer processor, network, and/or location to another) prevents manipulation of the video data without evidence of the manipulation because the video data committed to the blockchain ledger cannot be edited/modified/manipulated after the video data has been committed to the blockchain ledger. Thus, if the video data has been manipulated, a clear record has been formed and examination of the blockchain entries of the video data (and associated metadata) will show when (and by whom) the video data was manipulated.

Method 50 includes many of the same steps as set out in the method associated with system 10. However, method 50 includes the steps of committing the video data (and associated metadata) to the blockchain ledger regularly. Method 50 can include other steps not expressly disclosed herein, and the steps of method 50 can be altered and/or performed in any sequence without departing from the scope of the invention.

Step 52 includes collecting streaming video data by a camera or other streaming video source. The video data can be collected by, for example, camera 12. The collection of the streaming video data can be the creation of the video data by continuously recording a potentially ever-changing frame (with a field of view) by the camera. For example, camera 12 can record the inside of a warehouse in which consumer products are temporarily stored before being shipped out to one or multiple stores. The collection of streaming video data can be the recording of the movement of those products (and machinery used to move the products).

Step 54 includes committing the streaming video data and associated metadata to a blockchain ledger. Step 54 can be performed after the streaming video data (and associated metadata) has been collected/created, or step 54 can be performed at a time before the video data is conveyed/sent to a first computer processor (e.g., gateway 16). Alternatively, step 54 of committing the video data and metadata to the blockchain ledger can be performed at any other time

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before being conveyed/sent out from the camera. With the streaming video data and metadata being committed to the blockchain ledger prior to transmission of the video data and metadata from the camera (or from the network to which the camera is connected), a first chain/record of the video data and metadata is recorded in the blockchain ledger for later validation and authentication.

Committing the streaming video data (or first video data or altered first video data) to the blockchain ledger can include submitting and saving the streaming video data (and associated metadata) in multiple nodes of a distributed ledger (i.e., distributed ledgers). The distributed ledgers interact to confirm that the streaming video data submitted and saved is identical across all ledgers. Thus, the video data cannot practically be changed/manipulated after it has been committed to the blockchain ledger without evidence of such manipulation because the changes would need to be made to all of the identical entries of that streaming video data in each of the distributed ledgers. This characteristic of the blockchain ledger allows for commitment/recording of the video data to prevent manipulations/modifications of the video data.

Step 56 includes conveying/sending the streaming video data and metadata from the camera to the first computer processor, such as gateway 16. In other words, step 56 includes receiving the streaming video data and metadata by the first computer processor (e.g., gateway 16). As with system 10 above, camera 12 and gateway 16 can be, for example, distant from one another or physically or virtually adjacent to one another, such as being on the same network.

Step 58 includes committing the streaming video data and metadata to the blockchain ledger. After the streaming video data (and metadata) has been transmitted from the camera to the first computer processor (e.g., gateway 16), the streaming video data and metadata can be committed to the blockchain ledger to ensure that the streaming video data and metadata has not been manipulated between the time the streaming video data was collected by the camera and the time the streaming video data is received by the first computer processor. Manipulation of the video data can be identified, for example, by comparing the streaming video data and metadata recorded in the blockchain ledger in step 54 to the streaming video data and metadata recording in the blockchain ledger in step 58. If the two entries of streaming video data (and metadata) are different, then the streaming video data (and metadata) can be identified as altered or otherwise manipulated. If the two entries are identical, then the streaming video data (and metadata) can be identified as having not been altered or otherwise manipulated.

Step 60 includes preprocessing the streaming/incoming video data, by the first computer processor, according to preprocessing parameters defined within the configuration file (which can be edited/revised and updated at runtime). The preprocessing can include formatting the streaming/incoming video data to create first video data that is a subset of the streaming/incoming video data. For example, the first video data can have a first field of view that is less than a second field of view of the streaming/incoming video data collected by the camera. The preprocessing of the streaming video data according to the preprocessing parameters in the runtime configuration file to create the first video data can include other edits to the video data and the associated metadata as described above with regards to FIGS. 1-3.

Step 62 includes committing the first video data and metadata to the blockchain ledger. After the first video data (and associated metadata) is preprocessed, the first video data and metadata are committed to the blockchain ledger to

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record the edits that have been performed on the first video data (and metadata) by the first computer processor (e.g., gateway 16) according to the preprocessing parameters defined in the configuration file. Step 62, in committing the first video data and metadata to the blockchain ledger after the first video data has been preprocessed, ensures that if additional edits (e.g., manipulations) to the first video data (other than those performed prior to step 62) are later found when examining the first video data, it can be determined that those edits/manipulations were not performed by the first computer processor (e.g., gateway 16) or at the time that the first video data was committed to the blockchain ledger.

Step 64 includes publishing the first video data (and associated metadata) to the endpoint, which can be a proxy location and/or unified endpoint hosted by the first computer processor (e.g., gateway 16). The publishing in step 64 can be to/within an asynchronous messaging library, such as ZeroMQ. Further, the proxy location and/or unified endpoint to which the first video data (e.g., scenes 24A-24C) is published can be in wired or wireless communication with the first computer processor (e.g., gateway 16).

Step 66 includes accessing the first video data (e.g., the scene/topic) and associated metadata by the first subscriber (or any number of subscribers). The first subscriber can be a second computer processor or any other type of software and/or hardware. The preprocessed first video data can be edited/preprocessed for a desired use by the first subscriber, who then accesses the first video data (and associated metadata) potentially for further processing depending on the desired output/inference to be determined from the first video data. The first subscriber can be distant (e.g., physically remote) from the location at which the first video data is published such that step 66 can be performed physically and temporarily distant from step 64. For example, the first subscriber can be a second computer processor that is in communication with the endpoint via the internet, with the first video data being published to the endpoint by the first computer processor.

Step 68 includes again committing the first video data and metadata to the blockchain ledger. After the first video data (and metadata) has been transmitted from the first computer processor (e.g., gateway 16) to the first subscriber (e.g., the second computer processor and/or subscribers 26A-26D), the first video data and metadata can be committed to the blockchain ledger to ensure that the first video data and metadata has not been manipulated between the time the first video data was published by the first computer processor and the time the first video data is received by the first subscriber. Manipulation can be identified by comparing the first video data and metadata recorded in the blockchain ledger in step 62 to the first video data and metadata recording in the blockchain ledger in step 68. If the two entries of the first video data (and metadata) are different, then the first video data and metadata can be identified as altered/manipulated. If the two entries are identical, then the first video data and metadata can be identified as having not been altered/manipulated.

Step 70 includes processing the first video data, by the first subscriber (e.g., the second computer processor), to determine at least one output indicative of an inference dependent upon the first video data and/or to alter the first video data. For simplicity, the first video data (and associated metadata) that has been processed in step 70 can be referred to as altered first video data because the first video data can be changed/altered by the processing performed in step 70. Examples of the processing by the first subscriber are described above with regards to FIG. 5. Step 70 can

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include further editing of the first video data and/or meta-
data, which may change/alter the first video data as com-
pared to the first video data before step 70 (i.e., altered first
video data). Additionally, the processing of the first video
data in step 70 can include, for example, applying an AI
model to the first video data to determine and record at least
one output. In another example, optical character recognition
(OCR) is applied to the first video data to determine and
record any text that is shown in the first video data. Step 70
can include other processing to extract information/data
from the first video data in the form of one or more outputs.

Step 72 includes committing the altered first video data
and metadata to the blockchain ledger. After the altered first
video data (and associated metadata) is processed by the first
subscriber (e.g., the second computer processor), the altered
first video data and metadata are committed to the block-
chain ledger to record the edits/processing that have been
performed on the first video data (and metadata) by the first
subscriber (e.g., the second computer processor). Step 72, in
committing the altered first video data and metadata to the
blockchain ledger after the first video data has been pro-
cessed, ensures that if additional edits (e.g., manipulations)
to the altered first video data (other than those performed
prior to step 72) are later identified when examining the
altered first video data, it can be determined that those
edits/manipulations were not performed by the first sub-
scriber (e.g., the second computer processor).

Step 72 can also include committing any information
recorded during step 70 (during the processing by the first
subscriber) to the blockchain ledger. This information can
include, for example, any information/data created/collected
by an AI model (such as the amount of a product that passes
by in the first video data) and/or any text collected by OCR.
Committing the additional information created/collected/
recorded during step 70 provides a record of the informa-
tion/data that cannot be altered/modified once committed to
the blockchain ledger. Such information/data may be used
for proof of the particular output determined by the process-
ing, as well as validation and/or authentication of the infor-
mation and/or inferences derived from the information.

Step 74, which can be performed at any point during
method 50, includes examining the streaming video data, the
first video data, and/or the altered video data and associated
metadata as committed/recorded in the blockchain ledger.
Because the video data recorded in the blockchain ledger is
not able to be modified/altered, the video data is authentic
and used as proof of the state of the video data throughout
the collection, preprocessing, publishing, accessing, and
processing of that video data. If the video data was manipu-
lated at any point during method 50 (without notice and/or
authorization), the manipulated video data would be
recorded as is in the blockchain ledger and/or comparable to
a previous entry of the video data in the blockchain ledger.
Both situations, after step 74 of examining the video data
(and associated metadata) in the blockchain ledger, allow for
an identification of when and who manipulated the video
data without notice and/or authorization. Thus, method 50,
with multiple steps that include committing the video data to
the blockchain ledger, allows for validation of the authen-
ticity of the video data throughout the collection, preprocess-
ing, publishing, accessing, and processing of the video
data.

Step 74 can include comparing the streaming video data
(and associated metadata) as committed to the blockchain
ledger in step 54 and/or step 58 to the first video data (and
associated metadata) as committed to the blockchain ledger
in step 62 and/or step 68 and validating that the only

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difference between the streaming video data and the first
video data is the changes made during step 60, which is
preprocessing the streaming video data. Additionally, step 74
can include comparing the first video data (and associated
metadata) as committed to the blockchain ledger in step 62
and/or step 68 to the altered first video data (and associated
metadata) as committed to the blockchain ledger in step 72
and validating that the only difference between the first
video data and the altered video data is the changes made
during step 70, which is processing the first video data.
Moreover, step 74 can compare the streaming video data that
was committed to the blockchain ledger to the altered first
video data that was committed to the blockchain ledger in a
similar fashion to examine the differences and validate that
the only difference is the changes made to the streaming
video data during step 60 and step 70. If the differences
between the video data entries into the blockchain ledger are
other than the changes performed during step 60 and 70, it
can be determined that the video data was manipulated/
altered/modified without notice and/or authorization during
method 50. In such a situation, the video data would not be
validated as authentic.

FIG. 7 shows a system for blockchain recordation (i.e.,
commitment) and authentication of video data. Unified
endpoint 18 provides scene 24A (topic 1) having first video
data 21 to blockchain validation system 110 via providing
availability of scene 24A via publication at unified endpoint
18 (similar to system 10). To access scene 24A (video data
21), first AI model subscriber 126A, second AI model
subscriber 126B, and blockchain subscriber 126C subscribe
to topic 1 at unified endpoint 18. Each of first AI model
subscriber 126A, second AI model subscriber 126B, and/or
blockchain subscriber 126C can be one or multiple computer
processors at various locations near or distant from unified
endpoint 18 and each other. First AI model subscriber 126A,
second AI model subscriber 126B, and/or blockchain sub-
scriber 126C can be in communication with sink 176,
distributed ledger 178 (also referred to as blockchain ledger
178), and/or database 180.

First AI model subscriber 126A performs processing on
first video data 21 that is similar to (or the same as) AI model
48A, which is one of processing 48 that is performed on first
video data 21. Additionally, second AI model subscriber
126B performs processing on first video data 21 that is
similar to (or the same as) AI model 48A, but in system 110
first AI model subscriber 126A and second AI model 126B
can be analyzing/processing first video data 21 to determine
outputs that are indicative of different inferences. The deter-
mination of an output that is indicative of an inference can
include monitoring, by at least one of AI model subscribers
126A and 126B, video data 21 for the occurrence of the
inference and formulating the first output when the inference
occurs. Thus, the output is dependent upon the occurrence of
an inference.

When an inference is detected/determined, AI model
subscribers 126A and/or 126B can formulate the corre-
sponding outputs (e.g., the detection of an inference prompts
the formulation of an output). For example, first AI model
126A can be processing first video data 21 to determine an
output that is indicative of the amount of a specific product
that is viewable in scene 24A (i.e., a first inference), and
second AI model 126B can be processing first video data 21
to determine a different output that is indicative of how
many people appear in scene 24A over a defined period of
time (i.e., a second inference). The outputs by first AI model
subscriber 126A and second AI model subscriber 126B can
be annotated first video data and metadata (shown in FIG. 7

as first annotated video data **121A** and second annotated video data **121B**) that includes information detailing when, where, and why the inferences were determined by the AI models. This information can be included by annotating the video data and/or adding information to the metadata associated with the annotated video data.

For example, first annotated video data **121A** can show an outlined box overlaid over original video data **21** highlighting a specific product that is viewable in scene **24A** as well as a number overlaid over original video data **21** showing a probability that first AI model subscriber **126A** has correctly identified the specific product. The metadata associated with first annotated video data **121A** can include, in some examples, additional information related to the annotated video data, such as specifying the location on the frame of the video at which the specific product is viewable. Annotated video data **121A** and **121B** (and associated metadata) can include any other information used by and/or created by AI model subscribers **126A** and **126B**, respectively.

First AI model subscriber **126A** and second AI model subscriber **126B** can process video data **21** in parallel (as shown in FIG. 7) where each subscriber **126A** and **126B** subscribes to topic **1** to access video data **21** and each subscriber **126A** and **126B** performs processing/analysis on original video data **21** simultaneously to determine the occurrence of first inferences and second interferences associated with first AI model subscriber **126A** and second AI model subscriber **126B**, respectively. In another embodiment, first AI model subscriber **126A** and second AI model subscriber **126B** perform processing on video data **21** in series such that first AI model subscriber **126A** has access to video data **21** and processes video data **21** first and second AI model subscriber **126B** has access to annotated video data **121A** and processes annotated video data **121A** after it has been processed by first AI model subscriber **126A**, thereby outputting the information from both first AI model subscriber **126A** and second AI model subscriber **126B** as second annotated video data **121B**. While shown in FIG. 7 as only having two AI model subscribers **126A** and **126B** performing processing in parallel, other embodiments of system **110** can include any number of AI model subscribers performing processing in series and/or in parallel.

Blockchain subscriber **126C** accesses video data **21** by subscribing to topic **1** (i.e., scene **24A**). Blockchain subscriber **126C** can be configured to perform no processing of video data **21** and instead can just provide a relay (e.g., location) for original video data **21** for later use in commitment to distributed ledger **178** and database **180** along with first annotated video data **121A** and/or second annotated video data **121B**. Original video data **21** as accessed by blockchain subscriber **126C** may be useful in comparison to first annotated video data **121A** (which includes information added by first AI model subscriber **126A**) and/or second annotated video data **121B** (which includes information added by second AI model subscriber **126B**). System **110** can be configured to not include blockchain subscriber **126C**. For example, sink **176** could be configured to perform the functions of blockchain subscriber **126C** by subscribing to topic **1** at unified endpoint **18**. Additionally, blockchain subscriber **126C** can be combined with one or more other components of system **110**, such as being a part of sink **176**.

Sink **176** can include one or more controllers, computer processors, digital storage media, and/or other components that are in communication with first AI model subscriber **126A**, second AI model subscriber **126B**, and blockchain subscriber **126C**. Sink **176** can accept and store first annotated video data **121A** (and associated metadata), second

annotated video data **121B** (and associated metadata), and/or original first video data **21** and commit video data **121A**, **121B**, and/or **21** (and associated metadata) to distributed ledger **178** and/or database **180** when a trigger event occurs, which can be dependent upon the outputs/inferences determined by first AI model subscriber **126A** and/or second AI model subscriber **126B**. Alternatively, sink **176** does not need to store video data **121A**, **121B**, and/or **21** (and associated metadata) and can be a component that, when a trigger event occurs, performs instructions to commit video data **121A**, **121B**, and/or **21** (and associated metadata) to distributed ledger **178** and/or database **180**. Additionally, sink **176** can be a virtual location to which video data **121A**, **121B**, and **21** is relayed/sent/published until a trigger event prompts commitment to distributed ledger **178** and/or database **180**.

The event that triggers the commitment of video data **121A**, **121B**, and/or **21** to distributed ledger **178** and/or database **180** can be any event for which recordation and authentication is desired (e.g., proof of authenticity and/or that AI model subscribers **126A** and/or **126B** are functioning properly). The trigger event can be an instance (one frame of video data or a duration of time encompassing multiple frames of video data) when AI model subscribers **126A** and **126B** determine an output indicative of an inference. The inference can be anything of interest to AI model subscribers **126A** and/or **126B**. Some examples of an inference are as follows: a product that is viewable in video data **21** (i.e., scene **24A**), an incorrect product that is viewable in video data **21** (e.g., a product that is not the product of interest), a product that is viewable in video data **21** that is defective, a specified number of products that are viewable in video data **21** simultaneously or over a specified duration of time, any person that is viewable in video data **21**, any vehicles that are viewable in video data **21**, or anything else that is of interest to AI model subscribers **126A** and/or **126B**. In another example, video data **21** can show at least a portion of a manufacturing system used to construct a specific product and the inference occurs when the AI model determines that the specific product is incorrectly constructed (i.e., has a defect).

For ease of understanding, each commitment of video data **121A**, **121B**, and/or **21** (and associated metadata, which can include customer, business, and/or other related data) to distributed ledger **178** and/or database **180** is called an entry. Sink **176** can review video data **121A**, **121B**, and/or **21** for outputs indicative of inferences that are trigger events, or every inference determined by AI model subscribers **126A** and **126B** can be a trigger event that results in an entry being committed to blockchain ledger **178** and database **180**. Other embodiments/configurations can have first AI model subscriber **126A**, second AI model subscriber **126B**, and/or blockchain subscriber **126C** in communication with one another and configured to instruct one another to commit an entry to distributed ledger **178** and/or database **180** when each determines an output indicative of an inference that is a trigger event. In such a configuration, each subscriber **126A**, **126B**, and **126C** determines when one or multiple subscribers should provide video data to be part of an entry committed to distributed ledger **178** and/or database **180**. Further, system **110** can be configured such that the presence/determination of an inference by AI model subscribers **126A** and/or **126B** automatically triggers the commitment of an entry that encompasses that inference (along with original video data **21**) without the need for further instruction from sink **176** or other components.

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Each entry to distributed ledger **178** and/or database **180** can include video data from one, two, or all of video data **121A**, **121B**, and/or **21** (and associated metadata). For example, when first AI model subscriber **126A** determines an output that is indicative of an inference that warrants the commitment of an entry, both first annotated video data **121A** (from first AI model subscriber **126A**) and original video data **21** (from blockchain subscriber **126C**) can be committed as an entry to distributed ledger **178** and/or database **180** (without the committing of second annotated video data **121B**). In another situation, when first AI model subscriber **126A** determines an output that is indicative of an inference that warrants the commitment of an entry, all three of first annotated video data **121A**, second annotated video data **121B**, and original video data **21** can be committed as an entry to distributed ledger **178** and/or database **180**.

Each entry to distributed ledger **178** and/or database **180** can be one frame of specified video data (e.g., the frame/still image of video data that shows the inference determined by first AI model subscriber **126A** and/or second IA model subscriber **126B** as well as the one frame of original video data **21** of blockchain subscriber **126C**) or multiple frames/still images of the video data. For example, if first AI model subscriber **126A** determines an output that is indicative of an inference warranting commitment of an entry to distributed ledger **178** and/or database **180**, the entry can include ten frames of video data before the frame of interest (i.e., the frame that shows the inference), ten frames of video data after the frame of interest, and the frame of interest. Thus, the entry would include twenty-one frames of first annotated video data **121A**. Additionally, the entry could also include those same corresponding twenty-one frames of original video data **21** for comparison to first annotated video data **121A**. In such a configuration, with each entry including frames before and after the frame of interest (i.e., the frame showing the inference), it is likely that the inference (and associated metadata) will be captured and committed to distributed ledger **178** and/or database **180** for authenticity.

Other configurations can include entries with more or less frames per entry, and the number of frames can be dependent upon the inference that is determined/discovered by AI model subscribers **126A** and **126B**. For example, if the inference lasts a duration of time that is shown in forty frames, the entry to the distributed ledger **178** and/or database **180** can be fifty frames (which includes five frames before and five frames after the frames encompassing the inference).

Before being committed to distributed ledger **178** and/or database **180**, the entry can be assigned an identification number. The identification number can be saved in a storage media, such as database **180**, to enable identification and access to the entry as saved in distributed ledger **178**. The identification number should be the same number for the entry if it is committed/saved in multiple locations, and different entries should have different identification numbers than previous entries. Therefore, with each entry having a different identification number (and the number being saved in searchable database **180**), each entry is easily identifiable and locatable in distributed ledger **178**. The identification number can be assigned by/at sink **176** or by another component of system **110**, such as a computer processor.

Distributed ledger **178** can be a blockchain ledger that is public or private. Distributed ledger **178** can be a database with one or multiple storage media that allows for entries (that can include video data **121A**, **121B**, and/or **21** and associated metadata) to be committed/saved and shared across multiple networks, sites, institutions, geographies,

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processors, etc. (i.e., nodes). The participant at each node of distributed ledger **178** can access the entries committed/saved/shared across distributed ledger **178** and can own an identical copy of the entry. Any additions (i.e., new entries) to distributed ledger **178** are reflected and copied to all nodes. The nodes communicate to ensure that each entry is identically saved at each node. Thus, manipulation of the entries within distributed ledger **178** is difficult as each entry saved at each node would have to be manipulated without detection by distributed ledger **178**.

Sink **176** can include customer **184**, which can include a controller, computer processor, digital storage media, customer interface, and/or other components. Customer **184** can have access to all of the data coming to (and accessible by) sink **176**, such as first annotated AI video data **121A**, second annotated AI video data **121B**, and original video data **21** (and associated metadata). For example, the customer interface of customer **184** can be configured to view first annotated video data **121A**, second annotated video data **121B**, and/or original view data **21** (and associated metadata). Additionally, a node of distributed ledger **178** can be located at and/or under the control of customer **184** such that any information/entries committed to distributed ledger **178** is also accessible by customer **184**. Thus, customer **184** can evaluate and/or analyze video data **121A**, **121B**, and/or **21** and associated metadata to confirm/ensure the inferences are correct. In such a situation, because customer **184** is under the control of a node of distributed ledger **178**, customer **184** would have to authorize the commitment of the entry to distributed ledger **178**. Therefore, customer **184** can play a role in the confirmation of inferences and the commitment of entries to distributed ledger **178**.

Database **180** can include a computer processor, digital storage media, and/or other components that allowed for the organized collection of structured data, such as entries that includes video data **121A**, **121B**, and/or **21** and associated metadata (and potentially identification numbers for each entry). Database **180** can be modeled in rows and columns in a series of tables to make identifying and accessing the entries and the information associated with the entries (video data and/or metadata) easy and organized, though any type of database usable for storing the information in an organized manner for later retrieval can be utilized. Database **180** can be a separate component, or can be combined with sink **176**, blockchain subscriber **126C**, and/or other components of system **110**. Database **180** is configured to allow access and viewability of each entry without the need to access distributed ledger **178**. Thus, a party can look at the entries in database **180** as a preliminary analysis of the entry and then refer to the entry in distributed ledger **178** for an authenticated version of the entry. Because the entry can have the same identification number in database **180** as in distributed ledger **178**, the entry in distributed ledger **178** is easy to find as it corresponds to the entry in database **180**. Additionally, database **180** can include other information for identifying the entry in distributed ledger **178**, such as the date and time that the entry was committed to distributed ledger **178**. Further, database **180** can include or function in association with applications **182**, which can be configured to sort, analyze, edit, process, and/or otherwise examine entries in database **180** for any desired purpose.

The entries committed to distributed ledger **178** do not need verification from any other system, and can be standalone data that does not need information or reference to any other system. Thus, if the entries need to be accessed and analyzed long after being committed to distributed ledger **178**, no further information or context is needed by the

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analyzing party to fully understand and review video data **121A**, **121B**, and/or **21** (and associated metadata). The entry in distributed ledger **178**, even when accessed long after the entry is committed, is credible and authentic due to the characteristics of distributed ledger **178** and the commitment process of the entry (which includes video data **121A**, **121B**, and **21** (and associated metadata)) by system **110**. Thus, the authenticity of first annotated video data **121A**, second annotated video data **121B**, and/or original video data **21** can be confirmed by reviewing the entry of that video data in distributed ledger **178**.

System **110** can include other components and capabilities not expressly disclosed herein. For example, system **110** can have an audit system that reports data to customer **184** continuously or periodically. In one example, audit system can report to customer **184** (or other interested parties) a summary of inferences determined by AI model subscribers **126A** and **126B** and/or entries committed to distributed ledger **178**. In another example, audit system can report a list of the entries committed to distributed ledger **178** along with the identification number for each entry.

While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

The invention claimed is:

1. A method of committing video data and associated metadata to a blockchain ledger, the method comprising:
 - preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model;
 - accessing the preprocessed video data and associated metadata by the first artificial intelligence model, the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data;
 - formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference;
 - accessing the preprocessed video data and associated metadata by a blockchain subscriber independent from the first artificial intelligence model;
 - in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, committing a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data, wherein the blockchain ledger includes multiple nodes each having digital storage media and the entry is committed for storage at each node;
 - committing the first entry to a first node of the multiple nodes of the blockchain ledger, the first node being located at a customer terminal; and
 - accessing at least one of the first annotated video data, metadata associated with the first annotated video data,

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the preprocessed video data, and the metadata associated with the preprocessed video data at the customer terminal.

2. The method of claim 1, further comprising: assigning an identification number to the first entry.
3. The method of claim 2, further comprising: accessing the first entry in the blockchain ledger using the identification number.
4. The method of claim 3, further comprising: comparing the first annotated video data and metadata associated with the first annotated video data to the video data and metadata associated with the video data of the first entry in the blockchain ledger to confirm an authenticity of the first annotated video data and metadata associated with the first annotated video data.
5. The method of claim 1, further comprising:
 - in response to the determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, saving the first entry to a database that includes digital storage media.
6. The method of claim 5, further comprising: assigning an identification number to the first entry that is both committed to the blockchain ledger and saved to the database.
7. The method of claim 1, wherein the step of accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the video data comprises:
 - monitoring, by the first artificial intelligence model, the preprocessed video data for the occurrence of the first inference; and
 - formulating the first output in response to an occurrence of the first inference.
8. The method of claim 1, wherein the video data shows at least a portion of a manufacturing system for constructing a product and the first inference occurs in response to a determination by the first artificial intelligence model that the product is incorrectly constructed.
9. The method of claim 1, further comprising:
 - accessing the preprocessed video data and associated metadata by a second artificial intelligence model, the second artificial intelligence model being configured to determine a second output that is indicative of a second inference dependent upon the preprocessed video data;
 - formulating second annotated video data and associated metadata by the second artificial intelligence model including the second output indicative of the second inference; and
 - in response to a determination by the second artificial intelligence model that the second inference has occurred so as to prompt the formulation of the second output, committing a second entry to the blockchain ledger that includes the second annotated video data, metadata associated with the second annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data.
10. A system for authenticating video data and metadata associated with the video data, the system comprising:
 - a blockchain ledger;
 - a gateway, which includes a first computer processor, that is configured to preprocess the video data according to preprocessing parameters defined within a runtime configuration file to edit the video data to be suitable for use by a first artificial intelligence model;

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the first artificial intelligence model configured to:

access the preprocessed video data and associated metadata,

determine a first output that is indicative of a first inference dependent upon the preprocessed video data, and

formulate first annotated video data and associated metadata that includes the first output indicative of the first inference;

a blockchain subscriber, which is executed by a second computer processor, that is independent from the first artificial intelligence model and is configured to access the preprocessed video data and associated metadata;

a sink in communication with the first artificial intelligence model and the blockchain subscriber, the sink being configured to, in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, commit a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data; and

a second artificial intelligence model configured to access the video data and associated metadata, determine a second output that is indicative of a second inference dependent upon the video data, and formulate second annotated video data and associated metadata that includes the second output indicative of the second inference,

wherein the sink is in communication with the second artificial intelligence model and configured to, in response to a determination by the second artificial intelligence model that the second inference has occurred so as to prompt formulation of the second output, commit a second entry to the blockchain ledger that includes the second annotated video data, metadata associated with the second annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data.

11. The system of claim **10**, further comprising:

a database having digital storage media,

wherein the sink is configured to, in response to the determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, save the first entry to the database.

12. The system of claim **11**, further comprising:

a third computer processor in communication with the sink, the third computer processor being configured to assign an identification number to the first entry committed to the blockchain ledger and saved to the database.

13. The system of claim **11**, wherein the sink includes the third computer processor such that the sink assigns the identification number to the first entry.

14. The system of claim **11**, further comprising:

a fourth computer processor in communication with the database and configured to access the first entry in the database to view at least one of the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data.

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15. The system of claim **10**, further comprising:

a customer terminal able to access the first annotated video data and metadata associated with the first annotated video data.

16. The system of claim **10**, wherein the video data shows at least a portion of a manufacturing system for constructing a product and the first inference occurs in response to a determination by the first artificial intelligence model that the product is incorrectly constructed.

17. The system of claim **10**, wherein the first artificial intelligence model and the second artificial intelligence model access and analyze the video data simultaneously to determine the occurrence of the first inference and the second inference, respectively.

18. A method of committing video data and associated metadata to a blockchain ledger, the method comprising:

preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model;

accessing the preprocessed video data and associated metadata by the first artificial intelligence model, the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data;

formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference;

accessing the preprocessed video data and associated metadata by a blockchain subscriber independent from the first artificial intelligence model;

in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, committing a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data;

assigning an identification number to the first entry;

accessing the first entry in the blockchain ledger using the identification number; and

comparing the first annotated video data and metadata associated with the first annotated video data to the video data and metadata associated with the video data of the first entry in the blockchain ledger to confirm an authenticity of the first annotated video data and metadata associated with the first annotated video data.

19. The method of claim **18**, wherein the step of accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the video data comprises:

monitoring, by the first artificial intelligence model, the preprocessed video data for the occurrence of the first inference; and

formulating the first output in response to an occurrence of the first inference.

20. The method of claim **18**, further comprising:

in response to the determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, saving the first entry to a database that includes digital storage media.

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